

Topological polymer networks: a generative framework

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Abstract

The mechanical performance of crosslinked polymer networks is traditionally predicted by crosslink density, yet topologically diverse networks with identical densities exhibit vast property variations. Here, we introduce a generative computational framework that maps abstract graph topologies into physically equilibrated continuum networks. We validate this approach using atomistic molecular dynamics of polydimethylsiloxane (PDMS) and fluoropropyl-methylsiloxane (FPDMS). The models reproduce chemical specificity, capturing the distinct glass transition temperatures driven by side-group sterics while successfully erasing initial lattice artifacts. By extending the analysis to a coarse-grained ensemble of over 300 unique architectures at constant average degree (i.e., crosslinking density), we quantify the isolated role of topology in fracture mechanics. We identify a state-dependent hierarchy of design drivers: fracture in brittle networks is initiated by local stress concentrations (high betweenness centrality), whereas toughness in high-performance networks is governed by cooperative load

sharing (high eigenvector centrality). These results demonstrate that topological order serves as an orthogonal design variable to crosslink density, enabling the optimization of mechanical response without altering chemical composition.

Introduction

Predicting the thermomechanical performance of crosslinked polymer networks remains a fundamental challenge in materials science. The inherent disorder of the amorphous state obscures the direct relationship between synthesis conditions and macroscopic mechanical properties, creating a persistent structure-property gap.^{1,2} Consequently, the macroscopic response is determined not by mean-field averages, but by the specific network architecture established during curing.^{3,4} To accurately describe the network state, we require statistical descriptors that transcend simple averages. Specifically, while chain-length distributions ($P(N)$) dictate elasticity,⁵ the coupled crosslink-functionality ($P(f)$) and topological defect distributions govern non-linear failure properties, including tensile toughness and ultimate tensile strength (UTS),⁶⁻⁹ where bond rupture events are increasingly linked to shortest-path statistics and centrality metrics.^{6,10}

Historically, the field has relied on metrics such as the average crosslink density ($\langle \nu \rangle$) to describe the network state. However, detailed structural analysis demonstrates that this metric proves insufficient for capturing the true topological structure.¹¹ Critically, networks with nominally identical $\langle \nu \rangle$ exhibit significant scatter in their mechanical response solely due to variations in defect topology and spatial heterogeneity.¹²⁻¹⁴ The ongoing search for more robust descriptors, such as effective crosslink density, confirms the fundamental limitations of mean-field averages.¹⁵

Thus, establishing a rigorous structure-property relationship necessitates methods capable of direct, *a priori* control over the underlying network topology. Current algorithms for generating molecular dynamics (MD) structures do not address this requirement. Distance-based methods and kinetic Monte Carlo schemes¹⁶⁻¹⁸ simulate the crosslinking process di-

rectly. While physically intuitive, these approaches yield topology as an emergent, uncontrolled outcome rather than a design input,¹⁹ a limitation that persists even in modern covalent adaptable networks where topology remains largely stochastic.²⁰ Similarly, the current lattice-based approaches^{21,22} offer deterministic control but retain the crystallographic signatures of the underlying grid. This artificial anisotropy²¹ restricts their utility for modeling the random, isotropic topology characteristic of amorphous thermosets.

We present a multi-scale computational framework that prescribes network topology *a priori*, decoupling it from the stochastic curing path. Unlike kinetic approaches, we model the network as an explicit graph object before chemical instantiation, conceptually inverting traditional methods that map melts to meshes.²³ This allows for the direct manipulation of topological variables, such as the variance of the crosslink functionality ($P(f)$), as independent input parameters rather than emergent outputs. Notably, this allows us to apply the identical network structure to different chemical systems. This is impossible in kinetic simulations, where changing the chemical composition inevitably alters the resulting network topology. By bypassing the computational expense of reaction simulation, this method enables the rapid generation of extensive statistical ensembles. We exploit this capability to identify the specific graph-theoretical origins of mechanical property scatter, establishing network topology as a distinct, tunable dimension for the inverse design of soft materials.

We validate the physicochemical accuracy of the framework through atomistic MD simulations of polydimethylsiloxane (PDMS) and poly(3,3,3-trifluoropropylmethylsiloxane) (FPDMS) networks. Our analysis reveals that the steric hindrance of fluorinated side groups yields increased chain stiffness (scaling exponent $\nu \approx 0.48$) and elevated glass transition temperatures ($T_g \approx 200$ K) compared to PDMS, consistent with experimental expectations.^{24,25} Additionally, the local chain statistics exhibit a Brownian bridge distribution, confirming the presence of topological constraints. Extending this analysis to coarse-grained ensembles, we demonstrate that fracture strength and toughness depend critically on graph-theoretical metrics rather than the mean crosslink density. Notably, we observe that networks with high

eigenvector centrality exhibit enhanced toughness, as cooperative load-sharing architectures prevent stress localization.

We begin by describing the graph generation and embedding algorithms. Following this, we validate the chemical fidelity of the framework through an analysis of local chain statistics and thermodynamic transitions in atomistic networks. Finally, we use the scalability of the model to generate ensembles of coarse-grained networks to isolate the graph-theoretical origins of property scatter, focusing on metrics such as centrality and assortativity.

Methodology

To achieve systematic decoupling of topological architecture from chemical specificity, we employ a multi-stage computational framework. This approach deviates from traditional kinetic molecular dynamics (MD) schemes, which simulate the temporal evolution of crosslinking reactions. While kinetic methods offer physical realism, they treat the final network topology as an emergent, uncontrolled outcome of the stochastic trajectory.^{16,26–28} In contrast, we utilize a constructive graph-theoretical approach, enabling the direct, *a priori* prescription of the network’s connectivity map. This strategy allows for the precise specification of statistical descriptors (e.g., the degree distribution $P(f)$ and cycle rank), facilitating the inverse design of material properties.²⁹

Topological generation

The abstract network topology is generated using a novel constraint-satisfaction algorithm based on lattice pruning. The process is initialized by instantiating a fully connected precursor graph mapped onto a lattice scaffold (e.g., SC, BCC, or FCC) using only first-nearest-neighbor connectivity. The simulation domain dimensions (N_x, N_y, N_z) are independent, allowing for the construction of non-cubic geometries essential for studying aspect-ratio dependent phenomena in one- and two-dimensional systems, such as fibers and thin films. Fur-

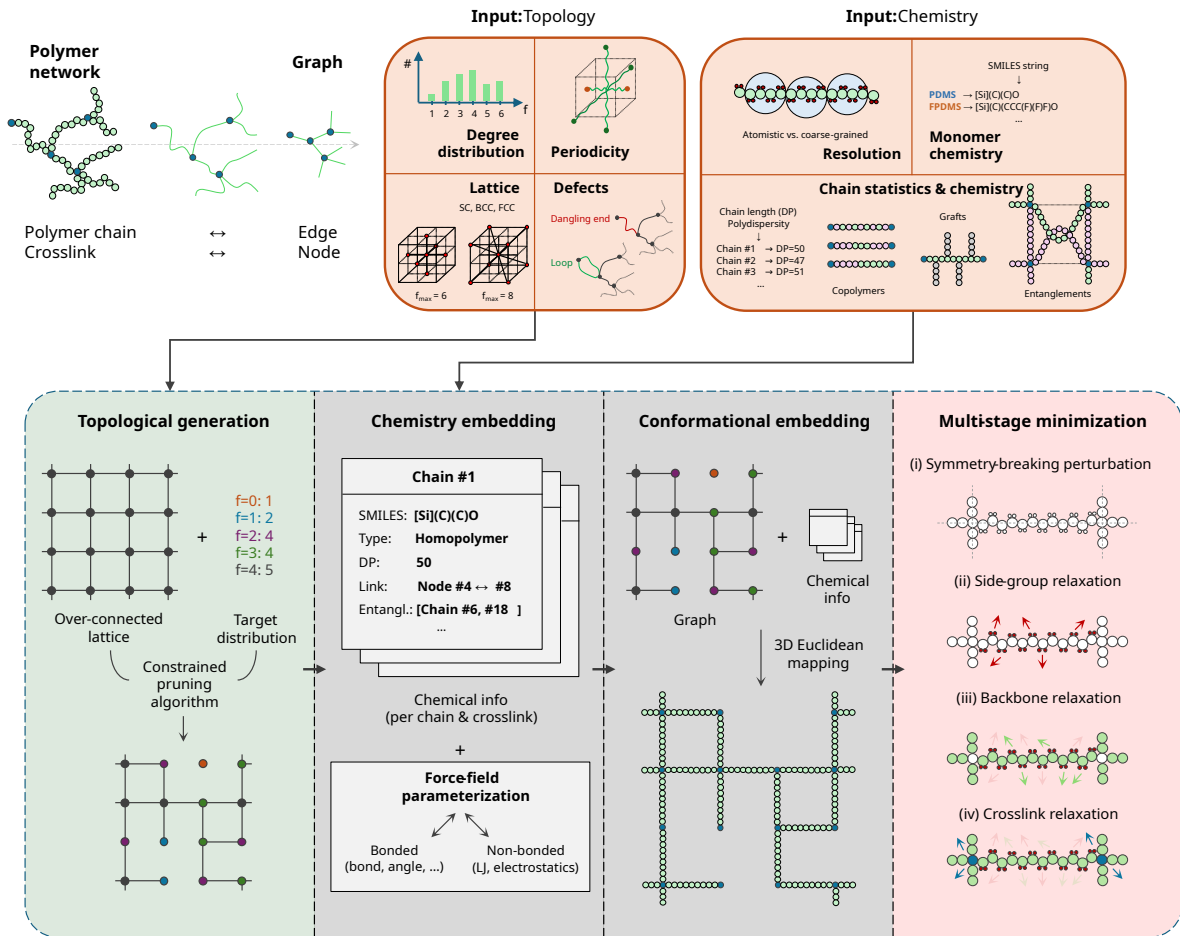


Figure 1: **Generative computational framework for topology-controlled polymer networks.** The workflow integrates abstract graph theory with physicochemical parameterization through a modular pipeline. **Input parameters** define the target architectural constraints (e.g., degree distribution, lattice periodicity) and chemical resolution (e.g., monomer SMILES, copolymer sequence, chain polydispersity). **Topological generation** employs a constrained subtractive pruning algorithm to carve a specific adjacency matrix from an over-connected lattice scaffold. **Chemistry embedding** instantiates each network chain as a discrete data object, assigning specific molecular identities and force-field parameters (bonded and non-bonded) to the topological edges. **Conformational embedding** maps the abstract graph into continuous 3D Euclidean space, translating connectivity into initial geometric coordinates. Finally, **Multistage minimization** executes a hierarchical relaxation protocol, comprising (i) symmetry-breaking perturbation, (ii) side-group relaxation, (iii) backbone relaxation, and (iv) crosslink relaxation, to resolve high-energy steric overlaps and generate a stable, clash-free configuration suitable for subsequent MD equilibration.

thermore, the framework accommodates both fully periodic boundaries for bulk systems and partially aperiodic constraints, extending its applicability beyond standard isotropic melts.

We emphasize that this lattice serves strictly as a high-density search space to guarantee efficient monomer packing; it does not impose a structural template, and its crystallographic symmetry is erased during the subsequent physicochemical embedding.³⁰

From this over-connected state ($f_{lattice} > f_{target}$), the target topology is sculpted via a hierarchical stochastic pruning process. Unlike random bond percolation, which typically yields uncontrolled, broad degree distributions dependent on a single probability parameter,³⁰ our novel algorithm removes edges iteratively to satisfy a complex, user-defined degree distribution $P(f)$ while strictly maintaining the Giant Connected Component (GCC),³¹ consistent with the connectivity thresholds established for soft random geometric graphs.³² However, we acknowledge that the realizability of a target $P(f)$ is non-trivial; not all mathematical distributions can be embedded within a Euclidean lattice due to geometric frustration and the constraints of the Handshake Lemma. As the lattice size increases, the combinatorial search space expands, revealing a topological degeneracy where multiple distinct architectures can satisfy a single statistical target. Consequently, while $P(f)$ alone is an insufficient descriptor of network topology, this degeneracy provides a vast library for inverse design, allowing specific architectures to be filtered from the ensemble based on targeted graph metrics. The pruning is executed in three distinct stages. First, the algorithm performs a targeted implantation of defects, where nodes are randomly selected to act as unreacted sites ($f = 0$) or dangling ends ($f = 1$) until the specific target densities are met. Second, it enforces chemical realism by iteratively removing edges from any node exceeding the maximum valency (f_{max}) of the target monomer (e.g., $f_{max} = 4$ for silicon centers). Third, a Monte Carlo minimization loop targets the remaining edges to converge the global $P(f)$ distribution to the input specification. Throughout this process, a Union-Find data structure validates global connectivity before every deletion, ensuring the final graph represents a single, cohesive network free of isolated clusters. Following this generation phase, the workflow supports a supplementary modification step to introduce higher-order topological defects. By scanning the established graph for permissible node or edge configurations, the algo-

rithm systematically inserts specific non-ideal motifs, such as loops, enabling the controlled study of network imperfections.

Chemical and conformational embedding

The translation of the abstract network into a physical model is performed via a deterministic embedding routine that separates connectivity from chemical identity. We employ a modular assignment routine to map molecular definitions to the graph elements. For atomistic models, the RDKit library³³ interprets monomeric SMILES strings to generate chemically heterogeneous networks. The framework automates force-field parameterization by recursively traversing the graph topology to identify and assign all required angle, dihedral, and improper potentials based on local chemical environments (i.e., neighboring atom types) and hybridization states. This ensures that the complex connectivity of the network is fully physically described without the need for manual topology definition.

This framework accommodates diverse molecular architectures, including chain-length polydispersity, sequence-defined copolymers (e.g., alternating or blocky), and discrete mixtures of chemically distinct chains. Furthermore, the protocol enables the systematic incorporation of side-chain grafts, parameterized directly by their monomeric SMILES, degree of polymerization, and grafting density. Although we parameterize these systems using the DREIDING force field³⁴ in the present work, the underlying mapping algorithm is force-field agnostic, theoretically supporting the integration of arbitrary interaction potentials. Similarly, for mesoscale investigations, the framework utilizes the standard Kremer-Grest bead-spring formalism,³⁵ mapping bead types directly to topological features without algorithmic modification.

Geometric instantiation maps the lattice nodes into 3D Euclidean space, generating polymer chains via interpolation along the minimum image vector to satisfy periodic boundary conditions. Unlike phantom network models, our approach explicitly accounts for topological constraints: we introduce algorithmic entanglements by modifying the embedding path

to create knots consistent with a user-defined entanglement density.³⁶ This capability allows the model to systematically capture the effects of the entanglement molecular weight (N_e) on mechanical response. Additionally, the protocol supports the instantiation of volumetric crosslinks (e.g., POSS cages),²⁶ enabling the simulation of high-functionality junctions ($f \gg 4$) and rigid spatial constraints distinct from standard point-like nodes.

Multi-staged relaxation and equilibration

The initial embedding of chains via linear interpolation results in a geometrically accurate but energetically unfavorable state, characterized by significant atomic overlaps. Furthermore, the lattice-based origin of these coordinates creates artificial symmetries that can stall standard gradient-based minimization algorithms due to vanishing forces. To resolve these conflicts without inducing numerical instability, we implement a multi-stage relaxation protocol. First, we apply a symmetry-breaking geometric perturbation. This routine iteratively identifies hard steric clashes and applies stochastic displacements to conflicting atoms, effectively resolving the most severe overlaps and providing the necessary directional bias for energy minimization.

Following this geometric resolution, the process proceeds to a serial, soft-potential minimization where degrees of freedom are released sequentially. We relax lighter atoms (e.g., hydrogens) and side groups first, followed by chain backbones, and finally the crosslink junctions; this hierarchical unfreezing prevents the distortion of the topological skeleton while mitigating high-energy interactions.¹⁶

Subsequently, the system transitions to dynamic time integration using a force-limiting algorithm. This integration scheme permits the gradual ramping of non-bonded potentials from soft repulsive barriers to the full interaction potential, safely converting residual potential energy into kinetic motion. At this stage, the simulation is fully stable under standard force-field constraints, allowing for unrestricted molecular dynamics and user-defined equilibration protocols. For the material properties reported in this work, we employed a

comprehensive thermal annealing cycle ($T \gg T_g$) to ensure the complete erasure of lattice correlations and the attainment of a relaxed, amorphous state.

Results and discussion

The utility of any generative algorithm depends on its ability to reproduce physical properties consistent with experimental reality. In this section, we first assess the physicochemical fidelity of our framework using fully atomistic simulations. We focus on two silicone variants, PDMS and FPDMS, which differ only in their side-group bulkiness. This comparison allows us to test if the generative framework respects the distinct spatial requirements and conformational statistics of different monomer chemistries. We specifically analyze the radial distribution functions and characteristic ratios to ensure the networks are amorphous and equilibrated.

Once the structural baseline is established, we shift our focus to the macroscopic mechanical response. Predicting stochastic failure properties, such as toughness, necessitates robust statistical sampling across diverse topological realizations. To achieve the necessary statistical power, we adopt a coarse-grained bead-spring representation for the fracture studies. This approach enables us to quantify the effect of topological descriptors on the stress-strain behavior, decoupling these geometric effects from the specific chemical identity of the chain backbone.

Structural characterization of atomistic networks

To validate the efficacy of our generative and relaxation framework in producing physically equilibrated amorphous networks, we perform a multiscale structural analysis. Figure 2a presents the radial distribution functions (RDF), $g(r)$, between crosslinking nodes in PDMS and FPDMS networks with strand length $N = 30$, compared against the initial crystalline reference lattice. The reference exhibits sharp Bragg peaks at characteristic lattice spacings,

while both networks show a complete absence of long-range order, with $g(r) \approx 1$ beyond the first coordination shell. This confirms that the relaxation protocol successfully eliminates structural correlations from the initial scaffold, yielding a homogeneous, amorphous network for both chemistries.

Local monomer packing is governed by chemistry rather than network architecture. Figure 2b displays the monomer–monomer RDF for PDMS across varying degrees of polymerization ($DP = 10, 30, 100$). The profiles overlap nearly exactly in the short-range regime ($r < 10$ Å), indicating that nearest-neighbor distances and local coordination are dictated by local bonded potentials and excluded volume, independent of strand length. The slight smoothing of higher- r features for $DP = 100$ arises from the broader conformational ensemble accessible to longer strands, reflecting increased configurational entropy rather than altered local structure.

At the mesoscale, the conformation of network strands reveals how chemical modification interacts with topological constraints. Figure 3a presents the scaling of the mean-squared end-to-end distance $\langle R_{ee}^2 \rangle$ with degree of polymerization N , following $\langle R_{ee}^2 \rangle \sim N^{2\nu}$. For PDMS, we find $\nu \approx 0.43$, indicating a contraction relative to the ideal Gaussian chain ($\nu = 0.5$) due to the combined effects of crosslink pinning and geometric confinement. In contrast, FPDMS exhibits $\nu \approx 0.48$, approaching ideal scaling. This expansion reflects the influence of bulky trifluoropropyl side groups, which increase steric repulsion and local chain stiffness, effectively counteracting network-induced collapse. The corresponding end-to-end distance distributions $P(R)$ in Figure 3b verify this observation, as for equivalent DP , FPDMS histograms are shifted toward larger R_{ee} values compared to PDMS, with both sets of distributions broadening systematically with increasing chain length.

Finally, we directly test whether the simulated strands obey the statistics of a crosslinked network. In a rubber, each strand is pinned at both ends (i.e., a Brownian bridge), which fundamentally alters its conformational statistics relative to a free chain.³⁷ Figure 4 shows the characteristic ratio $C_n = \langle R^2(n) \rangle / (n\ell^2)$ as a function of sub-chain index n for PDMS

and FPDMS at $DP = 10, 30, 100$. For all systems, C_n rises to an apparent plateau at small n before decaying significantly as n approaches the total strand length N . This decay is the signature of the bridge constraint, where the fixed chain ends suppress long-wavelength fluctuations, limiting the dimensions of large sub-chains.³⁷ Solid lines represent predictions from a semi-flexible Brownian bridge model, with the chain stiffness parameter C_∞ (representing intrinsic backbone rigidity) tuned to match the plateau region. While the model captures the general behavior, the simulation data decays more strictly at large n . This deviation arises because the theoretical model assumes a phantom chain, whereas the simulated strands are subject to quenched topological constraints; even in the absence of trapped entanglements, the non-crossability of neighboring chains creates a topological corridor that suppresses transverse fluctuations relative to the phantom prediction.³⁸

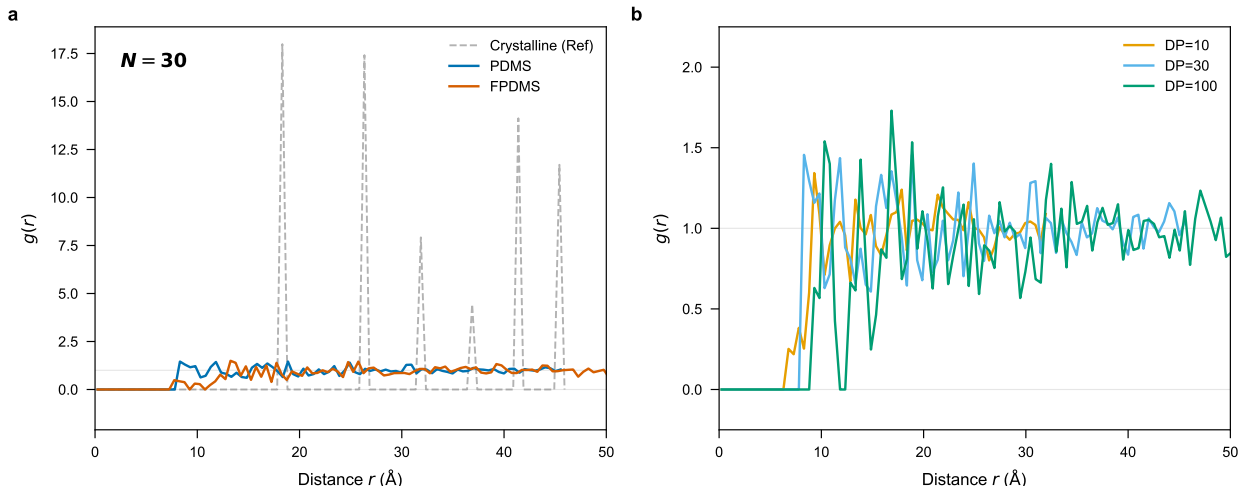


Figure 2: **Local structure and loss of lattice memory.** (a) Radial distribution function $g(r)$ between crosslink nodes for PDMS and FPDMS networks ($N = 30$) compared to the crystalline reference. The absence of Bragg peaks confirms complete amorphization. (b) Monomer-monomer RDF for PDMS at different strand lengths. Local packing is independent of DP , indicating chemistry-driven short-range order.

Thermodynamic validation

To assess the framework’s ability to capture chemical specificity beyond static structure, we characterize the glass transition temperature (T_g) of the equilibrated networks. We determine

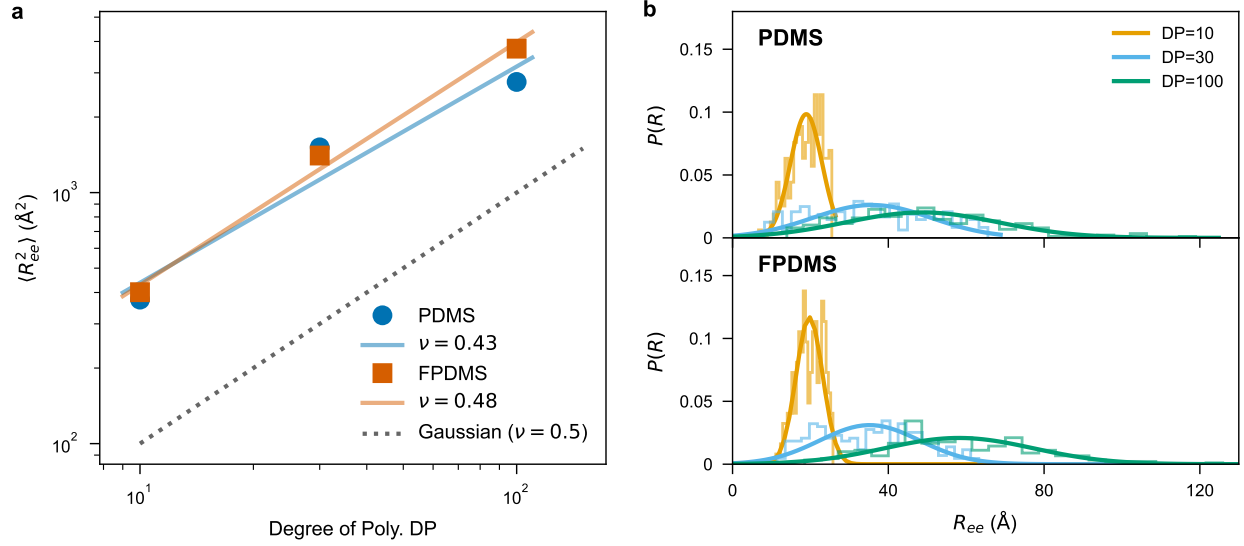


Figure 3: **Chain scaling and end-to-end distributions.** (a) Mean-squared end-to-end distance $\langle R_{ee}^2 \rangle$ versus degree of polymerization N for PDMS ($\nu \approx 0.43$) and FPDMS ($\nu \approx 0.48$). Dashed line indicates ideal Gaussian scaling ($\nu = 0.5$). (b) Probability distributions $P(R)$ of the end-to-end distance for $DP = 10, 30, 100$. FPDMS chains adopt more extended conformations than PDMS at the same DP .

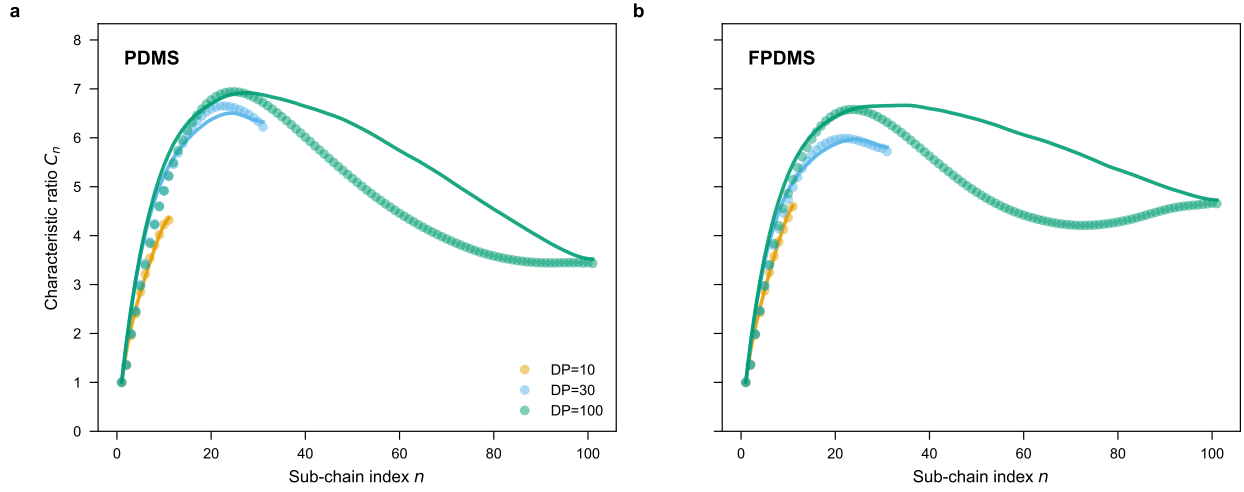


Figure 4: **Brownian bridge statistics of network strands.** Characteristic ratio C_n as a function of sub-chain index n for (a) PDMS and (b) FPDMS networks. Simulation data (symbols) show the characteristic decay as $n \rightarrow N$, in quantitative agreement with the theoretical semi-flexible Brownian-bridge model (solid lines). This confirms that strands are topologically pinned at crosslinks, validating the network's entropic elasticity.

T_g using a stepwise cooling protocol, sweeping from 310 K down to 100 K in 5 K decrements.

At each temperature step, the system is equilibrated for 0.2 ns before data collection. We

monitor the mean-squared displacement (MSD) of network atoms over a fixed short interval ($t = 4$ ps) to probe local segmental mobility. To reduce statistical noise, we report the average MSD calculated from five independent short trajectories at each temperature. We quantify T_g by fitting the temperature-dependent MSD data to a continuous piecewise bilinear function, defining the transition temperature as the intersection point between the vibrational (glassy) and mobile (rubbery) regimes.

Figure 5 compares the thermal response for PDMS and FPDMS networks instantiated from identical graph topologies. For PDMS, we obtain $T_g \approx 152$ K. This value aligns well with experimental benchmarks for crosslinked PDMS elastomers, which report transitions in the 146–154 K range depending on formulation.³⁹ For FPDMS, we observe a substantial increase to $T_g \approx 193$ K. This result is consistent with the experimental value of ≈ 203 K (-70°C) reported for fluorinated polysiloxanes.⁴⁰ These transition temperatures show minimal variation across different chain lengths ($DP = 10$ to 100), indicating that in these densely crosslinked networks, the glass transition is governed mainly by local segmental dynamics rather than global topology.

The observed ~ 40 K increase in T_g is a direct consequence of the distinct physicochemical properties of the trifluoropropyl side groups.⁴⁰ Fluorination introduces three key effects: (1) the higher atomic mass of fluorine atoms slows down segmental motion; (2) the strong C–F dipole moments increase intermolecular friction; and (3) the bulkier side groups reduce local free volume. Together, these factors significantly suppress local segmental mobility, raising the energy barrier for the onset of rubbery behavior. This ability to quantitatively reproduce both the absolute T_g values and the magnitude of their difference confirms that the force-field embedding and relaxation procedures successfully translate abstract chemical motifs into realistic, chemistry-specific dynamic behavior.

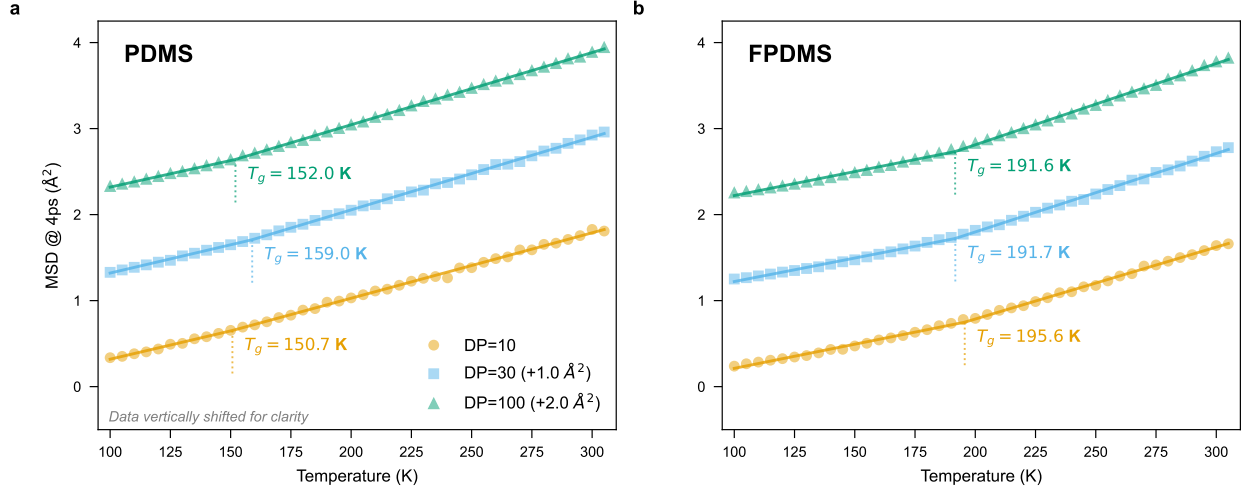


Figure 5: **Dynamic glass transition analysis.** Atomic mean-squared displacement (MSD) at 4 ps as a function of temperature for (a) PDMS and (b) FPDMS networks. The intersection of linear fits to the glassy and rubbery regimes defines T_g . FPDMS displays a significantly higher T_g (~ 193 K) compared to PDMS (~ 152 K) due to the steric hindrance of fluorinated side groups. Data series for different DP are vertically shifted for clarity; the transition temperature remains largely independent of strand length.

Data-driven topological mapping

To isolate the topological origins of mechanical failure, we generated an ensemble of over 300 unique network architectures. We constrained the average crosslink functionality to $\langle k \rangle \approx 3.5$ across the dataset but systematically varied the specific degree distributions $P(f)$ and connectivity patterns, including exactly 10 degree-0 ($f = 0$) nodes per system to ensure amorphous packing. We modeled these systems using a coarse-grained bead-spring formalism.³⁵ Bonded interactions were defined by a quartic potential (allowing for covalent scission), while non-bonded interactions utilized a Lennard-Jones potential with an attractive tail ($r_c = 2.5\sigma$) to reproduce the cohesion of a dense polymer melt ($\rho \approx 0.89\sigma^{-3}$). Following an equilibration protocol (10^7 steps, $\Delta t = 0.002\tau$), we subjected each network to uniaxial tensile deformation at a quasi-static strain rate of $\dot{\epsilon} = 5 \times 10^{-4}\tau^{-1}$. We averaged the stress-strain response over three orthogonal deformation axes to eliminate anisotropic artifacts.

Figure 6a maps the mechanical performance of the ensemble. Despite identical aver-

age crosslink densities, the networks exhibit a wide dispersion in ultimate tensile strength (UTS) and toughness (energy to break). We partition this domain into four quadrants based on median performance thresholds: Weak/Brittle (Q1), Weak/Tough (Q2), Strong/Brittle (Q3), and Strong/Tough (Q4). Analyzing the crosslink functionality distributions $P(f)$ for these groups reveals a clear structural signature (Figure 6b). Low-performance networks (Q1) exhibit broad distributions with a disproportionate fraction of high-functionality nodes ($f = 5, 6$). This skew indicates that brittle failure is driven by the presence of dense, high-degree clusters, which likely act as local stress concentrators.^{41–43} Notably, the transition to the intermediate toughness regime (Q2) coincides with a reduction in these degree-6 defects and the emergence of a bimodal distribution (peaks at $f = 3$ and 5). This suggests that while removing the most severe outliers enhances toughness, the resulting structural segregation into high- and low-connectivity domains prevents the material from achieving high strength. Conversely, high-performance networks (Q4) display a sharp, unimodal concentration of connectivity around the mean ($f = 3, 4$). This confirms that structural homogeneity is a prerequisite for mechanical synergy; eliminating both outliers and bimodality in favor of a uniform distribution minimizes weak links and maximizes cooperative load-bearing, consistent with the enhanced yield strength observed in homogeneously crosslinked elastomers.⁴⁴

To quantify which topological features drive the transition between these mechanical states, we calculated the standardized effect size (Cohen’s d) for various graph metrics. This statistic measures the difference in means between two groups normalized by their pooled standard deviation; values of $|d| > 0.5$ and $|d| > 0.8$ are typically considered medium and large effects, respectively. Figure 7a presents a heatmap of these drivers. We observe that the global stiffness (λ_2) acts as a universal promoter of mechanical integrity. We distinguish here between the standard algebraic connectivity (λ_2) and the backbone stiffness ($\lambda_{2,core}$). The latter is calculated on the graph’s k -core (where $k \geq 2$), obtained by recursively pruning dangling ends ($f = 1$ nodes) until convergence. While $\lambda_{2,core}$ isolates the connectivity of the load-bearing skeleton, our analysis reveals that the standard λ_2 is the significantly stronger

predictor of performance (Total $d = 1.4$ vs. 0.4). This dominance suggests that peripheral defects (dangling ends), often treated as trivial, actively degrade the network’s global stability; thus, a metric sensitive to the entire topology (λ_2) captures the true mechanical state better than one focused solely on the backbone.

However, other drivers are highly state-dependent. For initially brittle materials (Q1), the primary improvement mechanism is the removal of local defects such as bridges and stress concentrations (Figure 7b). We quantify stress concentrations using maximum betweenness centrality, which identifies nodes that serve as bridges for many shortest paths. High betweenness implies a bottleneck where force lines converge, serving as primary loci for bond rupture events,⁴⁵ as confirmed by recent shortest-path fracture topology analysis.¹⁰ Once the network achieves a baseline level of connectivity (Q3), the optimization strategy shifts. Further toughening is driven by metrics associated with global load sharing, specifically average eigenvector centrality. Unlike degree centrality (which counts neighbors), eigenvector centrality assigns high scores to nodes connected to other highly connected nodes. This metric identifies *short-cuts* for stress propagation, allowing the network to delocalize energy dissipation across a larger volume effectively.⁴⁶ This hierarchy implies that mechanical optimization requires a sequential strategy, prioritizing the elimination of fatal defects to ensure connectivity before maximizing topological order to facilitate stress delocalization.

While backbone connectivity and centrality metrics emerge as primary drivers, it is instructive to examine topological features that exerted minimal influence within this specific design space. Notably, loop density (cycle rank) exhibits a negligible effect size ($|d| < 0.1$) across the transitions in our heatmap, consistent with theoretical predictions for phantom networks where loops of order ≥ 3 are elastically neutral.⁴⁷ This contrasts with previous studies on polymer networks, where topological loops were identified as critical defects that can modulate fracture toughness by altering chain extensibility.⁴⁸ We attribute this discrepancy to the constraints of our generative algorithm: by fixing the average crosslink density at $\langle k \rangle \approx 3.5$, we implicitly limit the variance in loop statistics relative to the broader variations

in connectivity patterns. This suggests that while loops are critical in systems with variable density, in highly crosslinked, density-constrained networks, the global arrangement of the backbone ($\lambda_{2,core}$) becomes the dominant predictor of failure. This dominance of global connectivity over local loop statistics implies that for high-density resins, mechanical resilience depends less on specific local motifs and more on the uniformity of the stress-bearing skeleton.

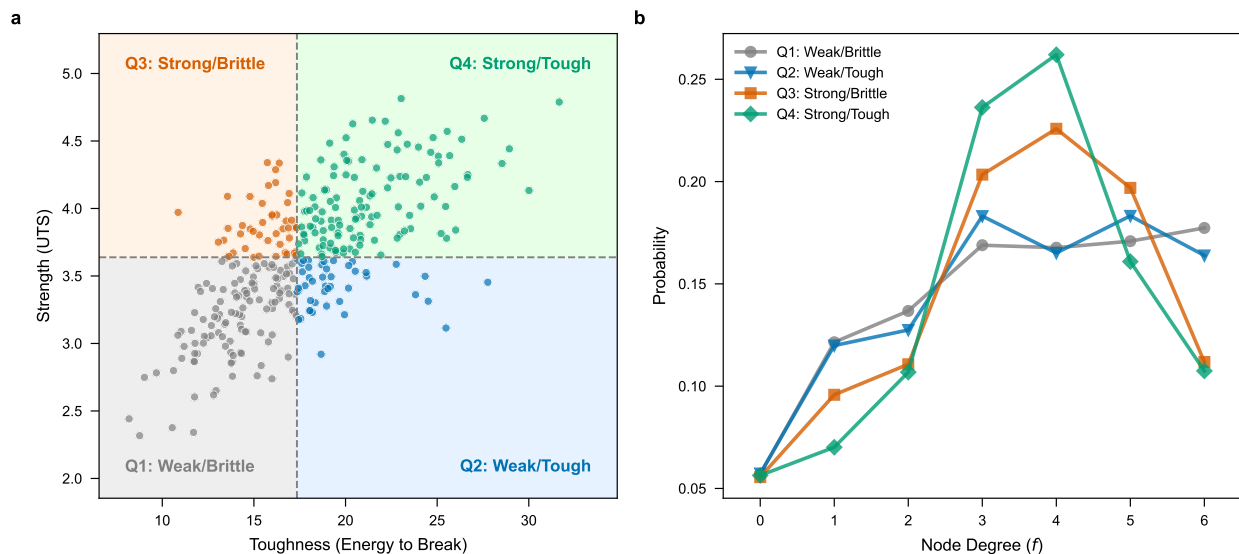


Figure 6: **Topological signatures of the strength-toughness synergy.** (a) Mechanical performance map of 300 coarse-grained networks with identical average connectivity ($\langle f \rangle \approx 3.5$). The design space is partitioned into four quadrants (Q1–Q4) based on median Strength and Toughness. (b) Node degree distributions $P(f)$ for each quadrant. High-performance Strong/Tough networks (Q4, green) exhibit a unimodal, homogenized distribution, whereas Weak/Brittle networks (Q1, gray) display significant structural heterogeneity.

Conclusion

In this work, we established a computational framework for the bottom-up design of polymer networks, bridging the gap between abstract graph theory and physical material properties. By implementing a constrained pruning algorithm followed by a multi-stage relaxation protocol, we demonstrated that lattice-based topologies can be mapped into continuous Euclidean space without retaining crystallographic artifacts. The modular nature of this workflow es-

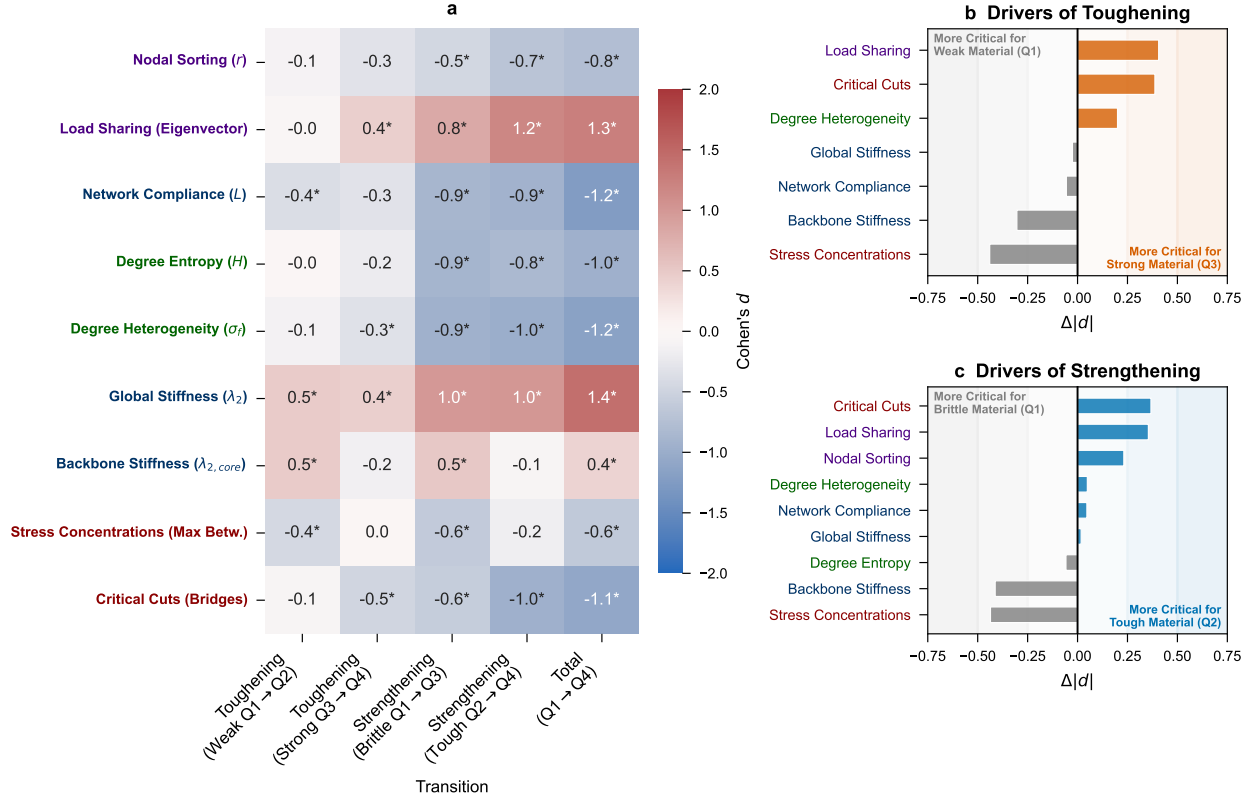


Figure 7: **State-dependent topological drivers.** (a) Heatmap of Cohen’s d effect sizes quantifying the impact of topological features on mechanical transitions. Red indicates a positive driver; blue indicates a penalty. (b) Hierarchy of toughening mechanisms. For weak materials (Q1), toughness is limited by local defects (Stress Concentrations). For strong materials (Q3), toughening is driven by global connectivity (Load Sharing). (c) Hierarchy of strengthening mechanisms, showing a similar shift from defect-sensitivity to topology-dependence.

establishes a foundation for generating diverse architectures beyond the specific systems analyzed here; the framework is readily extensible to polydisperse chain distributions, sequence-defined copolymers,⁴⁹ and entanglement constraints, while offering a pathway for modeling heterogeneous composite phases containing rigid-body inclusions (e.g., POSS nanocages or nanoparticles).^{26,50,51}

We validated the physicochemical fidelity of this approach through atomistic molecular dynamics of silicone elastomers. Structural analysis verified that the relaxation process erases the initial lattice memory, yielding isotropic radial distribution functions and chain conformations consistent with Brownian bridge statistics. Furthermore, thermody-

dynamic characterization confirmed that the embedding routine preserves chemical specificity. Despite sharing identical graph topologies, FPDMS networks exhibited a glass transition temperature approximately 40 K higher than their PDMS counterparts ($T_g \approx 193$ K vs. 152 K). This shift accurately reflects the increased steric hindrance and dipole interactions of fluorinated side groups, proving that the generative algorithm captures local segmental dynamics independent of the global connectivity.

Extending this analysis to the mesoscale, we utilized coarse-grained simulations to isolate the impact of network architecture on mechanical failure. By testing an ensemble of over 300 distinct topologies at constant crosslink density ($\langle f \rangle \approx 3.5$), we observed a significant dispersion in toughness and ultimate strength. Our analysis identified a state-dependent hierarchy of topological drivers. For brittle networks, mechanical performance is limited by local defects, specifically stress concentrations quantified by high betweenness centrality. In contrast, tough networks are governed by global connectivity metrics, such as eigenvector centrality, which facilitate cooperative load sharing. This transition highlights that optimizing a material requires different topological strategies depending on its baseline connectivity.

Finally, these results challenge the limits of classical rubber elasticity theory, which posits that macroscopic mechanical performance scales primarily with crosslink density (ν).⁵² By holding $\langle f \rangle$ (and thus ν) constant, we demonstrated that ultimate properties still vary significantly purely due to architectural heterogeneity. This decoupling suggests that topological design offers an orthogonal axis for material optimization, validating the distinction between density-driven elasticity and topology-driven failure established in model networks,⁴ while further allowing for the tuning of fracture mechanics without altering the fundamental chemical composition. This capability suggests a pathway for future materials design, where specific graph motifs are targeted to achieve distinct mechanical outcomes using chemically identical base materials, or vice versa.

Things to do:

- Add fiber network + percolation refs (Mikado, Broedersz) to Topological generation
- Preparation of GitHub page? TOPON: TOPological POLymer Networks / Algorithm?
- Verify title.
- Is E needed? (I also have a dataset where I use a slower deformation rate and strain until 0.2% to get Young modulus, no post-processing has been done on that yet.)

Acknowledgement

The authors acknowledge support from the Department of Mechanical Engineering at Northwestern University. We also thank the Northwestern University High Performance Computing Center for providing a supercomputing grant. This work was performed in collaboration with Dow.

Appendix: Definition of topological metrics

To quantify the structural architecture of the generated polymer networks, we utilize a set of eleven graph-theoretical descriptors derived from the adjacency and Laplacian matrices. This approach aligns with emerging frameworks that utilize graph-based descriptors to predict phase behaviors and structural properties in condensed matter systems.⁵³ The first family of metrics characterizes the global connectivity and spectral rigidity of the system. We define the global stiffness (λ_2) as the second smallest eigenvalue of the graph Laplacian, known as the algebraic connectivity. This value scales with the number of edges that must be cut to partition the graph, serving as a proxy for the overall mechanical resistance of the network. To isolate the mechanics of the load-bearing skeleton, we calculate the backbone stiffness ($\lambda_{2,core}$), defined as the algebraic connectivity of the 2-core subgraph obtained by recursively pruning dangling ends. We also evaluate the network compactness using the average shortest path length, which represents the mean number of steps required to traverse between any

pair of nodes; a lower value indicates a more compact topology capable of efficient global force transmission. Finally, we calculate the loop density (cycle rank), which counts the number of fundamental independent cycles in the graph ($M - N + 1$). This metric quantifies the abundance of topological loops, which is known to govern chain extensibility and the entropic elasticity of the rubber.

The second group of descriptors identifies local defects and bottlenecks that limit mechanical performance. We compute the stress concentrations using maximum betweenness centrality, which identifies the specific node that acts as a bridge for the highest fraction of shortest paths in the system. In a physical network, such nodes represent locations where force lines converge, creating spots likely to initiate failure. We also determine the critical cuts (bridges), defined as the fraction of edges in the network whose removal would increase the number of connected components. This metric quantifies the structural vulnerability of the network, representing the density of single bonds that are solely responsible for maintaining global connectivity.

The third family focuses on the statistical distribution of node connectivity (degree), characterizing the homogeneity of the crosslinking. We calculate the degree bimodality (D_{bi}) to quantify the separation between distinct populations of high-degree and low-degree nodes, indicating the presence of structural phase separation. We also measure the degree heterogeneity (σ_f), defined as the standard deviation of the degree distribution normalized by the mean degree. This coefficient of variation captures the breadth of the connectivity spread, distinguishing between uniform and polydisperse network architectures. Additionally, we compute the degree entropy (H) using the Shannon formulation, which measures the information content of the degree distribution. This serves as a proxy for the overall structural disorder, where higher entropy corresponds to a more randomized crosslinking pattern.

The final set of metrics captures higher-order correlations and cooperative mechanisms within the topology. We utilize the load sharing metric, defined as the average eigenvector

centrality, to measure the extent to which nodes are connected to other highly connected nodes. Unlike simple degree centrality, this recursive definition identifies the shortcuts of the network that facilitate the delocalization of stress across a large volume. Lastly, we calculate the assortativity coefficient (r), which quantifies the Pearson correlation between the degrees of connected nodes.⁵⁴ A positive assortativity indicates a segregated structure where hubs cluster together, while a negative assortativity implies a mixing where high-degree sites are surrounded by low-degree neighbors.

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